HDOC DAY 47

PROGRAM

Write a C program that reads the size and elements of a sorted variable-length integer array, arr[], in ascending order. Also, read an integer target. Pass the array to a function named ceil_index() using its address. Use pointers to pass the array between the calling function and ceil_index(). The function ceil_index() must:

- Find the index of the smallest element in arr[] that is ≥ target.
- Treat this element as the ceil of target.
- If multiple occurrences of the ceil element exist, return the index of its first occurrence.
- Display the ceil element and its index.
- Display -1 if no element is ≥ target.

CODE

```
#include <stdio.h>


int ceil_index(int *arr, int n, int target)
{
   int i;


   for (i = 0; i < n; i++)
   {
      if (*(arr + i) >= target)
      {
         return i;
      }
   }
```

```
    return -1;
}

int main()
{
    int n, i, target, index;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid array size.\n");
        return 1;
    }

    int arr[n];

    printf("Enter the sorted array elements in ascending order:\n");

    for (i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    printf("Enter target: ");
    scanf("%d", &target);
```

```
    index = ceil_index(arr, n, target);

    if (index != -1)
    {
        printf("Ceil element = %d\n", *(arr + index));
        printf("Index = %d\n", index);
    }
    else
    {
        printf("-1\n");
    }

    return 0;
}
```